

CSD 420 Advanced Java Programming

Module 2 Programming Assignment - Binary I/O

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Program Execution and Output

The screenshot below demonstrates the full workflow of both programs. DataWriter.java and DataReader.java are compiled together with no errors. DataWriter is then run twice in succession, producing two independent sessions of five random integers and five random doubles appended to contreras datafile.dat. DataReader then reads the file back and correctly displays both sessions in the order they were written, confirming that append mode preserves prior data. The final ls -la listing verifies the data file exists at 120 bytes, which matches the expected size of two sessions storing five 4-byte integers and five 8-byte doubles each.

```
(base) fernandocontreras@Fernandos-MacBook-Pro module-2 % ls
DataReader.java DataWriter.java

(base) fernandocontreras@Fernandos-MacBook-Pro module-2 % javac DataWriter.java DataReader.java

(base) fernandocontreras@Fernandos-MacBook-Pro module-2 % java DataWriter
java DataWriter
Writing integers to contreras datafile.dat:
82 6 17 75 58
Writing doubles to contreras datafile.dat:
89.4035 74.9214 18.3479 83.6573 74.2359

Data successfully written.
Writing integers to contreras datafile.dat:
41 29 32 13 6
Writing doubles to contreras datafile.dat:
54.4119 22.5098 18.7015 35.2815 18.0780

Data successfully written.

(base) fernandocontreras@Fernandos-MacBook-Pro module-2 % java DataReader
Reading data from contreras datafile.dat:

Session 1 integers:
82 6 17 75 58
Session 1 doubles:
89.4035 74.9214 18.3479 83.6573 74.2359

Session 2 integers:
41 29 32 13 6
Session 2 doubles:
54.4119 22.5098 18.7015 35.2815 18.0780

Session 3 integers:
Finished reading. Total sessions found: 2

(base) fernandocontreras@Fernandos-MacBook-Pro module-2 % ls -la
total 40
drwxr-xr-x@ 7 fernandocontreras  staff   224 Aug 20 15:41 .
drwxr-xr-x@ 9 fernandocontreras  staff   288 Aug 20 15:38 ..
-rw-r--r--@ 1 fernandocontreras  staff   120 Aug 20 15:41 contreras datafile.dat
-rw-r--r--@ 1 fernandocontreras  staff  2416 Aug 20 15:41 DataReader.class
-rw-r--r--@ 1 fernandocontreras  staff  2726 Aug 20 15:40 DataReader.java
-rw-r--r--@ 1 fernandocontreras  staff  2450 Aug 20 15:41 DataWriter.class
-rw-r--r--@ 1 fernandocontreras  staff  2652 Aug 20 15:40 DataWriter.java
```